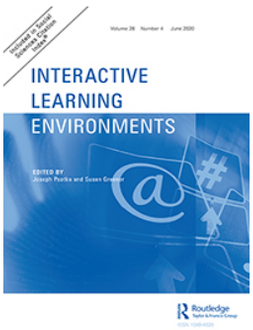

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Evaluation of AR embedded physical puzzle game on students' learning achievement and motivation on elementary natural science

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ABSTRACT

Emerging technologies have been widely used in applications designed to increase potential for learning enhancement. This study develops a mobile application to integrate augmented reality (AR) functionality into physical puzzle-type games. In the application, learners see visualizations of three-dimensional (3D) dinosaurs and interact with relevant multimedia learning content. To evaluate the effectiveness of the proposed platform, experiments were conducted to evaluate impact on learner achievement and motivation. Experimental (AR-based) and control (paper-based) groups included a total of 53 fifth-grade students. Pre- and post-activity questionnaires (adapted from the Attention, Relevance, Confidence, Satisfaction (ARCS) Model of Motivational Design), and pre- and post-learning quizzes were used for assessment. Results showed that the AR-based group scored significantly higher ($p < 0.05$) than the Paper-based group, raising implications for a new approach to teaching natural science at the elementary school level. However, the AR-based group also showed a relative drop in confidence, suggesting that further investigation is needed to understand the impact of AR technology on learner motivation. This study provides a case for the integration of AR technology into typical physical games. The results raise important concerns for further research in technology enhancement in education. Future research should include a wider range of content topics and target groups.

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Augmented reality; AR-embedded physical puzzle game; elementary natural science learning; ARCS model of motivational design; Cretaceous dinosaurs and fossils

Introduction

Game concepts have been widely applied in education to increase learner interest, motivation, and knowledge retention, using both physical games (Amory, Naicker, Vincent, & Adams, 1999; Dempsey, Rasmussen, & Lucassen, 1994; Hogle, 1996; Schwartzman, 1997) and digital games (Clark, Tanner-Smith, & Killingsworth, 2015; Connolly, Boyle, MacArthur, Hainey, & Boyle, 2012; Vlachopoulos & Makri, 2017). Such games typically consist of (physical or digital) boards, cards or puzzles that are used for drill and practice, warm-up activities or rewards (Charlton, Williams, & McLaughlin, 2005; Kapp, 2012). Typical game themes include adventure, competition, cooperation, or puzzle (Amory, 2001; Battocchi et al., 2010; Dempsey, Lucassen, Gilley, & Rasmussen, 1993; Jacobs & Dempsey, 1993; Pivec, 2007). Furthermore, games provide advantages including interactivity, pleasure, and flexibility for adaptive levels of difficulty or number of players (Shute & Ke, 2012).

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Among the various games, physical puzzles (made with wood or paper) are used as a tool to assist learning (Davis, Shepherd, & Zwiefelhofer, 2009). Physical puzzle-type games (e.g. jigsaw) require learners to reason and/or interact to solve a puzzle. Educational puzzle games typically have players assemble scrambled pieces or components into a correct order (Salen & Zimmerman, 2004). Such puzzles are commonly used by educators or practitioners to enhance the learning value of educational activities to enhance performance in a range of learning objectives, e.g. knowledge acquisition, application, or analysis.

New technologies are rapidly being integrated into games and learning materials, and game-based learning has been widely applied in classrooms (Chen, Wang, & Lin, 2015; Denham, Mayben, & Boman, 2016; Kafai, 2006). Augmented reality (AR) technology overlays virtual objects or information ("components") over real-world settings (Craig, 2013). These virtual components appear and disappear in smartphones or other handheld devices in response to certain triggers, thus providing additional context or activities related to the real-world surroundings (Azuma, 1997; Azuma et al., 2001; Milgram & Kishino, 1994). AR has unique advantages for creating immersive learning materials which offer supplementary digital learning contents overlaying a physical context. The use of AR and gamification (Laine, 2018; Hwang, Wu, Chen, & Tu, 2016; Kiryakova, Angelova, & Yordanova, 2014) provides a new possibility of engaging users in learning activities.

Previous studies have found the use of AR in educations to promote the development of problem solving and critical thinking skills, and to enhance learning motivation (Bower, Howe, McCredie, Robinson, & Grover, 2014; Cai, Chiang, Sun, Lin, & Lee, 2017; Hwang et al., 2016; Lu & Liu, 2015; Lu, Liu, & Wang, 2013; Santos et al., 2014). The use of AR in puzzle games allows for the inclusion of additional virtual content (e.g. 3D objects, and animations) associated with physical puzzle components.

This paper describes the design and evaluation of two versions of a learning game for fifth grade natural science students; one version is paper-based and the other integrates AR technology. The relative impacts on learning achievement and motivation of the two versions are investigated as addressed in the following research questions.

- Do learners using the AR-embedded physical puzzle game outperform those using a traditional paper-based puzzle game?
- Do learners using the AR-embedded physical puzzle game show improved learning motivation over those using the paper-based puzzle game?

The remainder of this paper is organized as follows: Section two reviews related work. Section three describes the design of the puzzle games and the development of the related learning content. Section four presents the experimental design and methods of evaluation. Sections five and six respectively present and discuss the experimental results. Section seven presents conclusions, research limitations and further directions.

Literature review

The use of AR allows for virtual objects associated with physical puzzle components. For example, AR embedded toys, i.e. match cards, flash cards, and jigsaw puzzles have been developed to teach everyday topics including animals, vegetables, vehicles, number, and colours, effectively improving learning performance in preschool children (Yilmaz, 2016). New toys incorporate virtual objects such as 3D objects and animations. Also, AR technology could potentially be used to guide players to better complete physical puzzle games. Lin et al. (2016) focused on the development of AR-based support to help learning-disabled individuals autonomously learn Chinese characters through tangrams. Users demonstrated improvement in completing puzzle games when compared with those using traditional paper-based learning.

Studies have also integrated AR for learning support in attempts to enhance traditional learning materials, such as embedding AR components in an electric circuit built on a breadboard (Alakärppä, Jaakkola, Väyrynen, & Häkkilä, 2017) or physical items from nature (Avilés & Cruz, 2017). AR has been applied to enhance learning by blending physical elements and digital content in ways which cannot be replicated by traditional paper-based solutions. While such approaches can enhance learning outcomes, the integration of AR into education applications raises issues related to usability, cognitive load, and technical proficiency (Akcayir & Akcayir, 2017). While AR concepts have already been integrated into physical puzzle games, research into such applications is scarce, and a contrast with learning outcomes from traditional physical puzzle games has yet to be developed.

Study design

From a problem-solving perspective, puzzles require users to assemble randomly scattered puzzle elements in a logical or predetermined manner. This entails cognitive activities including observation, guessing, attempting, exploring possibilities, and strategy development. Puzzle games feature a wide range of designs, including variables such as shape, size, and material type. Shape types include jigsaw, square, triangle, or circle. Puzzle sizes can be adjusted to accommodate different numbers of players, various learning objectives, and learners' physical capabilities. Puzzle elements can be made of various materials including paper, wood, or plastic, which contribute to ease of production, low cost and ease of use.

This study follows the six game design characteristics proposed by Garris, Ahlers, and Driskell (2002): fantasy, rules/goals, sensory stimuli, challenge, mystery, and control. We focus on the design of two puzzle games in terms of these characteristics to enhance learning effectiveness. The development of the two games requires consideration of appropriate puzzle shape, size, and material. The two puzzle games required learners to manipulate puzzle components on a grid-shaped board. The AR-based puzzle game included new design elements such as interaction with digital contents and the introduction of virtual characters. The virtual characters were designed to promote learning motivation in a story-centric theme (Lester, Rowe, & Mott, 2013). In the educational design stage (McKenney & Reeves, 2012), the development and implementation of physical objects and the mobile application were tested iteratively to ensure the design was usable and effective in actual learning settings. The paper-based group was seen as having an advantage in that, given limited resources, the instructor could autonomously develop supplementary learning tools (e.g. puzzles). However, additional ICT development resources, e.g. AR-embedded technology, would further enhance learning outcome. Therefore, the two types of learning games were developed and compared.

Physical puzzle game

Players were provided with a complete puzzle set and class handouts (Figure 1(a)). The puzzle set included twelve triangular and seven hexagonal puzzle pieces (Figure 1(b)). Class handouts include images of Cretaceous dinosaurs and their fossils. Players are asked to read the handout

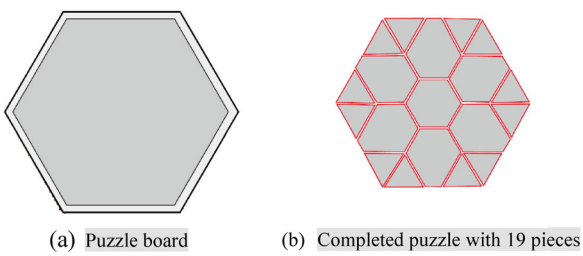


Figure 1. The design of the puzzle set.

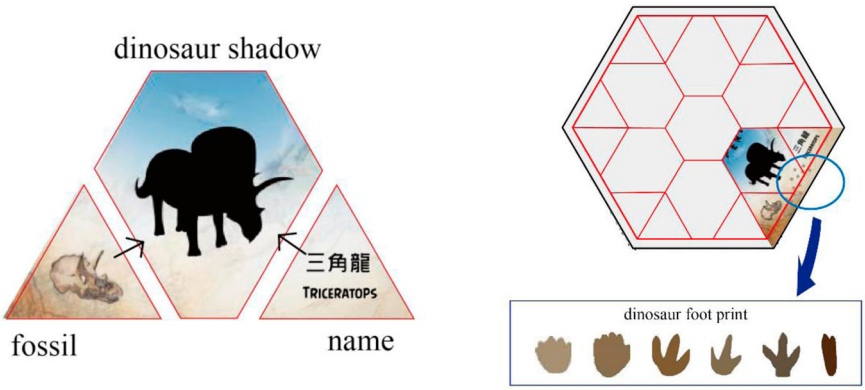
content aloud prior to the game. Each hexagonal puzzle piece depicts a silhouette of a dinosaur, while half of the triangular pieces show a drawing of a fossil, and the other half shows the Chinese and English names of a particular dinosaur. Learners were asked to assemble the 19 puzzle pieces on the board in such a way that each dinosaur silhouette was accompanied by its associated triangular pieces (see Figure 2(a)). Related pieces featured distinctive sets of dinosaur footprints, providing association clues (Figure 2(b)). As shown in Figure 3, the central hexagon puzzle piece displays a pair of boy and girl scouts who serve as virtual guides throughout the AR environment.

AR-embedded puzzle game

Learners in the experimental group used the same physical puzzle set, along with a digital application in a handheld device (tablet). The experimental group did not receive the handouts. The application supported puzzle theme completion by providing a 3D visualization of each dinosaur, along with multimedia learning content. The application is implemented in Android smart phones and tablets using Unity-Multiplatform. 3D digital models were built using Autodesk 3D Max. The tool to implement AR was developed by D'fusion.

Learners using the AR-based application needed to verify their solution correctness (see Figure 4), thus the application can check learner solutions/pairings and provide instantaneous feedback (see Figure 4(a–c)). Positive feedback is provided through a colourful 3D dinosaur icon (the control characteristic). The playing scenario allows AR-users to verify puzzle piece placement (in Figure 4(e) the colourful dinosaur icon indicates proper placement, whereas Figure 4(b) shows a grey dinosaur to indicate an error). The design aims to encourage learners to explore the digital learning content, with puzzle completion conducted through trial and error.

As shown in Figure 5, the learner activates the application in a 7"Android tablet. The camera is used to scan the real-world puzzle piece (Figure 5(b)). The scanned image of the dinosaur silhouette (Figure 5(c)) triggers a 3D model in grey (Figure 5(f)) on the screen. The learner can then rotate the image in three axes. Learners selects the associated fossil to change the dinosaur's colour and release a trophy – in this case a broken dinosaur egg (Figure 5(e)). Digital learning content is provided to help the learner determine the correct match. Once all six eggs were obtained, the learner was awarded a gold trophy (Figure 5(g)).



(a) Each set of three pieces included a hexagonal piece showing a dinosaur silhouette, and two triangular pieces showing a fossil and the dinosaur's name in Chinese and English.

(b) Distinctive footprints provided visual continuity clues for piece alignment.

Figure 2. Theme based arrangement in a puzzle set.

Figure 3. Six themes in total in the puzzle set.

Learning contents

Learning content was taken from the Science and Life Technology elementary school curriculum unit for rocks and minerals, which contained auxiliary materials on fossils. Content instruction took eighty minutes, using content modified from the websites of the National Taiwan Museum, Natural Taiwan Science Museum, and related textbooks and encyclopaedia illustrations. Six visually-distinct dinosaur types were selected: Triceratops, Ankylosaurs, Pachycephalosaurus, Tyrannosaurus, Quetzalcoatlus, and Therizinosaurus. The learning goal was to help learners understand the names, characteristics, lifestyles and fossil appearances of the different dinosaurs.

Learning content was developed based on the lower three layers of the conceptual framework of Bloom's revised taxonomy (Anderson & Krathwohl, 2001): remember, understand, and apply. For the remembering layer, we designed materials to help learners memorize dinosaur appearances, names, and key fossil parts. To support the understanding layer, we developed the evolutionary formations and definitions of fossils and life habits of the dinosaurs. For knowledge application, learners were required to understand differences in the periods in which the various dinosaurs lived and how

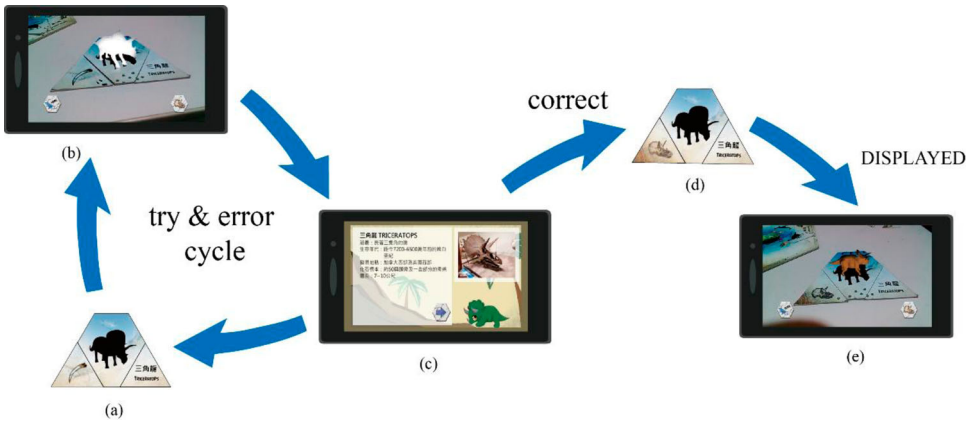


Figure 4. AR-based puzzle game showing two matching scenarios: (a) an uncomplete or wrong combination of puzzles; (b) a complete match of the puzzles.

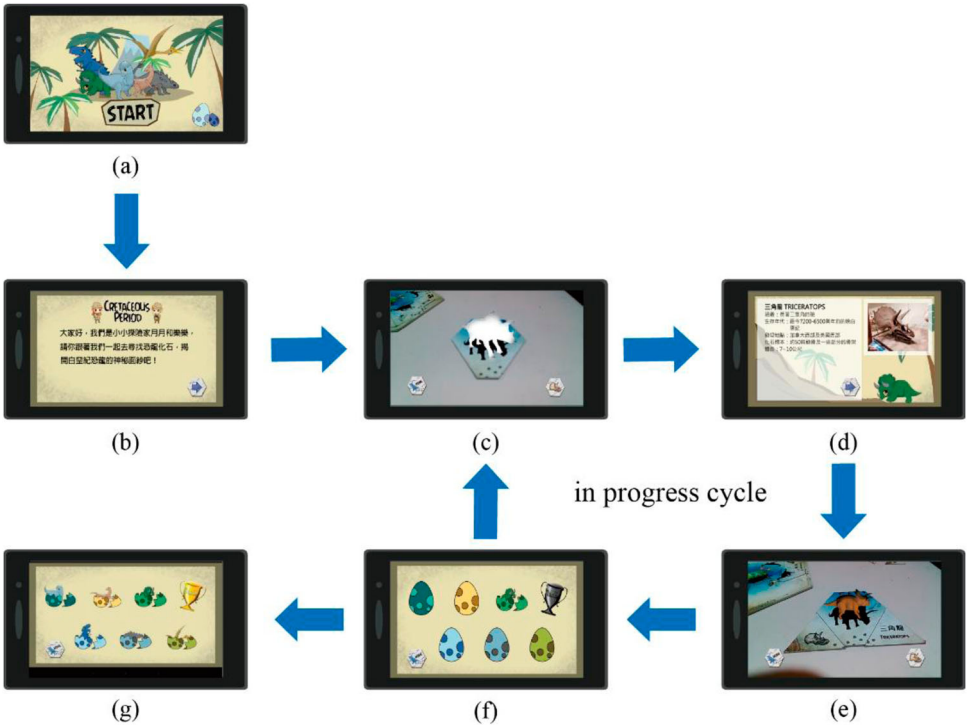


Figure 5. AR-based puzzle game showing the completing steps: (a) start off the content; (b) introduce the game; (c) display one dinosaur; (d) display the learning content; (e) complete one of the six set; (f) demonstrate all sets completed; (g) win the trophy.

the fossils were formed. The learning contents were reviewed and modified by two senior subject matter experts in natural science, one classroom teacher, and one university scholar.

Handout materials and learning contents of the application

The handouts for the paper-based puzzles were printed in colour on six A4 pages. Each page introduced a specific dinosaur through images and text. The images included a photo of the representative fossil skull and an illustration of the living dinosaur. Texts included the dinosaur's name (in Chinese and English) and a Chinese explanation of the name's meaning, the period in which the dinosaur lived, its height, and the number of such fossils found to date. Application-based learning content mirrored that of the paper-based handout.

Participants

The research was conducted using a pretest-postest quasi-experimental design. A total of 64 participants were drawn from two fifth grade classes at Long Pu Elementary School, New Taipei City, Taiwan. A total of nine students were excluded due to repeated absence, leaving a total of 53 participants completed the study (for a completion rate of 83%). Twenty-five subjects (13 male and 12 female) were assigned to the AR-based group, while twenty-eight subjects (16 male and 12 female) were in the Paper-based group (see Table 1).

Experimental procedures

The experiment lasted 80 min (see Figure 6), and consisted of a pre-test, followed by a verbal introduction, game play, and post-test. In the pre-test phase, each learner was given 15 min to complete

Table 1. The characteristics of the participants.

		AR-based ($n = 25$)		Paper-based ($n = 28$)		
Variables	Classification	*N*	%	*N*	%	*p*-value
Gender	Male	13	52.00%	16	57.14%	0.910
	Female	12	48.00%	12	42.86%	
Pre-test knowledge	Score	80.20 ± 3.67		81.21 ± 3.57		0.844

the motivation questionnaire and the written pre-test. Following the pre-test, the instructor and four teaching assistants divided each group into subgroups of five or six learners for a 20 min PowerPoint presentation of the lesson contents.

Following this introduction, each group was given 25 min to complete the game. Each team in the control group received a puzzle set and the A4 size handout, while each team in the experimental

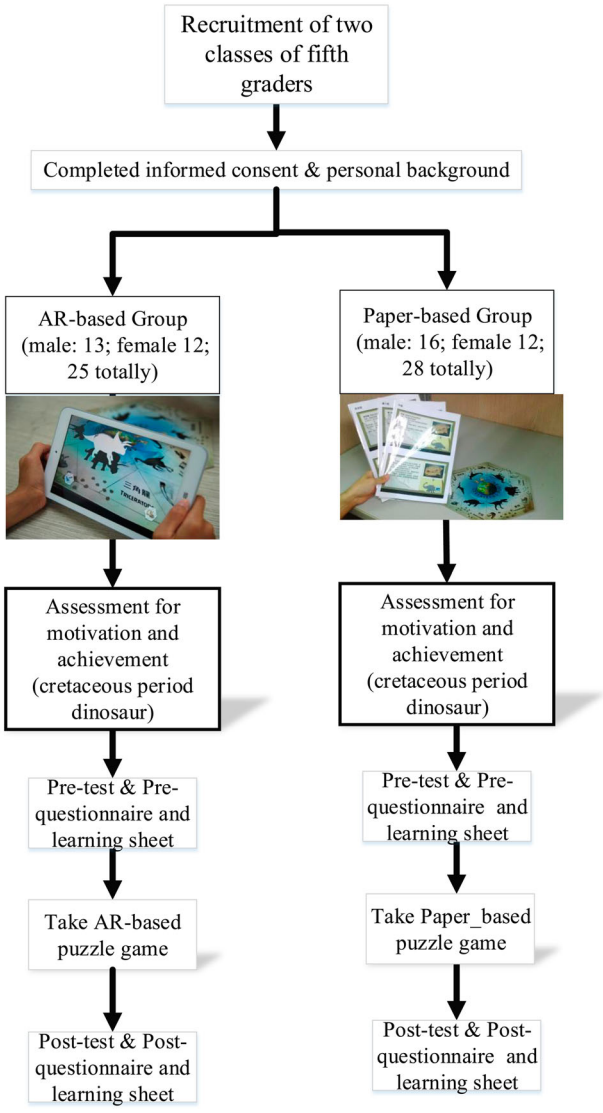


Figure 6. The flowchart of the evaluation.

group received a puzzle set and a 7-inch tablet. In each team, one member was the player designated to manipulate the puzzle pieces, while the other team members provide support. The player began the game by selecting a hexagonal piece at random.

In the control group, players reviewed the handout and then matched the hexagonal pieces with their associated triangular pieces through iterative trial-and-error. The players compared the triangular fossil components to select a match based on appearance with support from the handout. This process was then repeated with the dinosaur name components to complete the set. Each player browsed all the fossil puzzles, and then selected a potential fossil puzzle piece for comparison with the live dinosaur piece to determine similarities. Teaching assistants provided necessary clues for teams having trouble with the task. Once a player completed all six sets, the puzzles and the handout were passed to another student to restart the game.

Learners in the experimental group were also given 25 min to explore the Android-based application. The team player used the tablet camera to scan a hexagonal puzzle piece, producing a 3D, grey-coloured model on the display. Support could be retrieved from the associated textual lesson content, e.g. fossil information and dinosaur name. The player then selects a fossil piece as a possible match. Once the dinosaur has been matched correctly with its fossil and name, the colour of the model changes. Once a player completes all six sets, the tablet and the puzzle set were passed to another student to restart the game.

Once all students had completed the puzzle, they were given a post-activity motivation questionnaire and a written post-test slip which lasted 15 min.

Evaluation outcomes

The relative impact of the two puzzle types was assessed in terms of learning achievement and motivation. Learning achievement was assessed to evaluate knowledge improvement based on a comparison of pre-and post-test scores for each group, and a comparison of post-test scores for the two groups. Each pre-test and post-test had 20 items with a full score of 100 points. The quiz was designed based on the table of specifications (TOS) (Fives & DiDonato-Barnes, 2013) using Bloom's taxonomy to evaluate learners' overall understanding of the appearance, behaviour, footprints, and fossils of the various dinosaurs. Motivation was assessed using a questionnaire adapted from the ARCS Model of Motivation (Keller, 2010), including 24 questions related to Attention, Relevance, Confidence, and Satisfaction. The questionnaire's Cronbach's alpha (0.942) indicated high internal consistency and reliability.

The pair *t*-test was applied to compare pre–post performance within the two groups, while an independent *t*-test was used to compare pre- and post-test performance for the two groups. All statistical tests were two-tailed, and a *p*-value less than 0.05 indicated statistical significance. Statistical analysis was performed with SPSS v.20.0 (IBM Corp., Armonk, NY, USA).

Results

Participants

The baseline information distributions (Table 1) revealed no significant differences between the two groups in terms of gender or pre-test knowledge.

The baseline information distributions (Table 1) revealed no significant differences between the AR-based group and the paper-based group in terms of gender or pre-test knowledge.

Learning achievement

As shown in Table 2, scores increased significantly between the pre-test and the post-test for both the AR-based group ($p < 0.001$) and the Paper-based group ($p < 0.05$). Furthermore, a comparison of the

Table 2. Comparison of the pre-test and post-test ARCS learning motivation.

	AR-based ($n = 25$)	p-value	Paper-based ($n = 28$)	p-value
Pre-test	80.20 ± 3.37	0.000	81.21 ± 3.57	0.016
Post-test	96.00 ± 1.65		88.43 ± 2.31	

Table 3. Comparison of pre-test and post-test knowledge score.

		AR-based ($n = 25$)		Paper-based ($n = 28$)		
	Classification	N	%	N	%	p-value
Post-test	Score	96.00 ± 1.65		88.43 ± 2.31		0.012

post-test knowledge scores between the two groups shows the AR-based group scored significantly higher ($p < 0.05$) than the Paper-based group (Table 3).

Learning motivation

Based on Keller's (2010) ARCS Model of Motivation, Table 4 shows a significant drop in confidence among the AR-based group following the activity ($p < 0.05$), accompanied by slight decreases in attention and relevance. The ARCS categories in satisfaction slightly increase in the AR-based group. The Paper-based group showed a small but insignificant decrease in attention, relevance, and confidence, while slightly increase in satisfaction.

Discussion

The reported results provide a better understanding of the relative merits of AR-based and physical puzzle sets in terms of knowledge acquisition, learning motivation, and learning retention.

Learning achievement

As shown in Table 2, both groups showed significant improvement in knowledge acquisition. This indicated that both game types effectively promote enhanced learning. Similar improvements to learning achievement have been shown in mathematics learning (Hung et al., 2014), everyday

Table 4. Comparison of pre-test and post-test knowledge score.

		AR-based ($n = 25$)			Paper-based ($n = 28$)		
Category		Mean	SD	p-value	Mean	SD	p-value
Overall							
	Pretest	4.17	(0.58)	0.279	4.03	(0.65)	0.743
	Posttest	4.04	(0.62)		3.99	(0.56)	
Attention							
	Pretest	4.45	(0.55)	0.279	4.25	(0.53)	0.292
	Posttest	4.29	(0.70)		4.13	(0.67)	
Relevance							
	Pretest	3.99	(0.71)	0.468	3.89	(0.70)	0.851
	Posttest	3.89	(0.74)		3.86	(0.66)	
Confidence							
	Pretest	4.15	(0.60)	0.026	3.99	(0.76)	0.871
	Posttest	3.81	(0.59)		3.96	(0.62)	
Satisfaction							
	Pretest	4.13	(0.76)	0.527	4.01	(0.83)	0.901
	Posttest	4.22	(0.71)		4.03	(0.71)	

topics (Yilmaz, 2016), and Chinese characters (Lin et al., 2016). As shown in Table 3, the AR-based group outperformed ($p < 0.05$) the paper-based group on the post-test and the high post-test result of the AR group (over 96) suggests that technology-integrated physical puzzles may enhance some learning achievement, e.g. short term memory testing is enhanced in comparison with activities using typical puzzles. This is possibly due to the interactive support provided by the application functioning real-time feedback for correct and incorrect puzzle piece matching. In contrast, physical puzzle users might not immediately determine the correct answer from the handout and thus require additional support from group members, who may provide incorrect answers. Instructors observed that the paper-based groups tended to solve the puzzle faster, possibly because the physical puzzle format was conducive to improvised division of labour. On the other hand, puzzle solving in the AR group was a more solitary exercise, with one student using the tablet, while others either looked on or were not engaged.

Previous related studies had not given an indication of the impact of AR technology on learning achievement in comparison with typical physical puzzle games. Future investigation is suggested to identify factors that contribute to the enhancement effect of AR technology on learning outcomes.

Learning motivation

As shown in Table 4, only the AR group showed a significant decline in learner confidence, indicating that AR-group members might feel less confident following the activity and found further learning to be difficult. The significant reduction in confidence among the AR-based group was consistent with the findings of Cheng and Tsai (2013) which suggested that such activities might increase cognitive loading due to material volume and the need to familiarize oneself with learning processes for handheld devices, despite the otherwise intuitive nature of the assigned task. Furthermore, our experimental results contrasted with the findings of Hwang et al. (2016) as well as Chiang, Yang, and Hwang (2014) who noted that learning attitudes were further improved by AR-based approaches as students would see the introduction of AR application technology as fun and engaging. However, their comparison was between different types of mobile learning application (AR-based vs. typical), while the current study compares AR puzzles with traditional physical puzzles, which may explain the discrepancy. The review of previous AR studies (Akcayir & Akcayir, 2017) provided no indication of whether AR-based approaches produced a relative improvement to learning motivation, which required further investigations. From a learner-centric perspective, a holistic understanding of learner background is critical to ensuring positive learning experiences (Molin, 2017; Mooney, 2013), particularly in terms of learner familiarity with technology, instructional design, participant background (e.g. high or low achievers), or individual learning styles.

Another issue with both groups showed a slight decline in attention and relevance. Surprisingly, the paper-based group using physical puzzle sets alone failed to enhance learner attention, but this may possibly be due to the fact that such approaches are already widely used in classroom instructional activities and thus learner did not find the activity to be particularly novel. The AR-based group using augmented reality with tablet computers also failed to enhance learner attention, possibly because children already have extensive access to handheld devices, and have already been widely exposed to illustrations of dinosaurs. Also, many participants were considered to be higher achievers (with pre-test knowledge scores of 80 and above), indicating a high degree of familiarity with the learning contents, and this familiarity is likely to reduce learning challenge and engagement. The slight reduction in attention and relevance for both groups suggests a need to improve learning instructions as a means of enhancing self-reflection among participants. Another possible limitation of our research was that both groups had to share a puzzle game set, with up to six learners in the AR-based group taking turns using a single device, and group members may have suffered reduced engagement once their turn as player was over. Previous studies (Cruickshank & Telfer, 1980; Ritzko & Robinson, 2006) addressed the nature of learning games limits the number of participants in order to actively participate with the learning process and to minimize the noise of private

conversations. In addition, both group showed a slight increase in learner satisfaction, indicating that AR- and paper-based learnings contribute to a sense of engagement and utility.

Further investigation is suggested to determine whether device sharing reduces learning motivation and how it could be enhanced by group collaboration or discussion. Moreover, interacting with technology also produced unexpected learner behaviours not found in the paper-based group. For instance, some students attempted to overlay the AR image of one dinosaur over the card-based silhouette of another to simulate fighting between the two animals. Future work could explore the potential impact on learning of these types of spontaneous interactions. In addition, motivation could be related to competence. Therefore, factors that would affect learning achievement or motivation could be included, such as time for task completion and cognitive loading.

Conclusion

Augmented reality technology was integrated into a puzzle-type game, and its impact on learning outcomes was assessed in comparison to traditional physical puzzles in the context of an elementary school natural science lesson. Results indicated that both types of puzzle games enhance specific knowledge in natural science. The AR-based group had an advantage in terms of post-test results.

Results also showed that the use of physical puzzle games (both AR-enabled and traditional) did not have a significant impact on learning motivation for the fifth-grade students, although such tools are frequently used to promote learning. This raises important new implications for considering variables in participant's prior knowledge, puzzle game design, familiarity with ICT, and appropriate number of game set for learning. Producing consistently satisfactory learning outcomes is challenging, and the use of motivating aids such as emerging technologies or physical games in learning activities requires further investigation.

Due to the small sample size, care should be taken in generalizing the results of using the proposed activity design to other user groups. Further research should account for a wider range of content topics, target groups, psychological development stages, and geographic regions.

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Disclosure statement

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